YEAR 7 · UNIT 1 · HOMEWORK 2

Signs, Multiples and Cells

Mathematics 1.2 — Multiplying & Dividing Integers · 1.3 — Multiples and the LCM
Science 1.2 — Plant & Animal Cells · 1.3 — Specialised Cells

Name: ____________________________ **Class:** ____________________________

Date given: ______________________ **Date due:** _________________________

How to do this homework

- Write your answers **on this sheet**, in the spaces given. If you need more room, use the back of the page.
- In maths, **write the calculation before the answer**.
- In science, answer in **full English sentences**. "Haemoglobin" is not a sentence; "A red blood cell is full of haemoglobin" is.
- If a word is new to you, look for the blue **Word help** box on that page before you give up.

Section A · Mathematics 1.2 — Multiplying and Dividing Integers

Remember from the lesson

Same signs give a positive. Different signs give a negative. The same four rules work for $\times$ and for $\div$, because dividing undoes multiplying.

A **product** is the answer when you multiply. The product of 2 and -9 is -18.

A1 The four rules

(a) Fill in the sign that each rule gives you.

$+ \times + =$ _____ $+ \times - =$ _____ $- \times + =$ _____ $- \times - =$ _____

(b) Now work these out.

(a) $-6 \times 4 =$ _______ (d) $-36 \div 6 =$ _______ (g) $-9 \times 0 =$ _______

(b) $7 \times -3 =$ _______ (e) $45 \div -9 =$ _______ (h) $-12 \div -12 =$ _______

(c) $-5 \times -8 =$ _______ (f) $-56 \div -7 =$ _______ (i) $8 \times -7 =$ _______

(j) $-3 \times -3 =$ _______ (k) $-100 \div 4 =$ _______ (l) $-60 \div -5 =$ _______

A2 Say it, then write it

Word help — the English decides which number comes first

multiply -7 by 6	you **start** at -7: -7×6.
divide -40 by 8	you start at -40: $-40 \div 8$.
5 divided into -35	the same trap as last week — it is $-35 \div 5$, not $5 \div -35$.
the product of	the answer when you **multiply**.
times	the everyday English word for multiply.
share equally between	divide.

Turn each sentence into a calculation, then work out the answer. **Write the calculation first.**

1. Multiply -7 by 6.
   Calculation: ______________________________ Answer: ___________

2. Divide -40 by 8.
   Calculation: ______________________________ Answer: ___________

3. What is the product of -9 and -4?
   Calculation: ______________________________ Answer: ___________

4. 5 divided into -35.
   Calculation: ______________________________ Answer: ___________

5. Mr Bowen shares a debt of 24 dollars equally between 6 people. Write each person's share as a negative number.
   Calculation: ______________________________ Answer: ___________

A3 A sentence that is only half true

Think carefully about this one

People often say: **"Two negatives make a positive."**

It is a useful sentence, but it is only true *sometimes*. Test it yourself before you decide.

(a) Work out $\boxed{-3 \times -4}$ Answer: ___________

(b) Work out $\boxed{-3 + -4}$ Answer: ___________

(c) Is "two negatives make a positive" true for **both** of them? Answer in one full sentence.

(d) Rewrite the sentence so that it is **always** true. You will need to say which operations it works for.

A4 Brackets, and checking yourself

Two rules that save marks

Do the brackets first. Work out what is inside, then multiply or divide. Careful: if the bracket holds an **addition**, the sign rules above do **not** apply inside it.

Estimate: round each number to something easy, calculate, and check your real answer is close to it.

(a) Work these out. Show what the bracket comes to first.

(i) $20 \div (-3 + -2) =$ ____________

(ii) $(-4 + 10) \times -3 =$ ____________

(iii) $-6 \times (2 - 9) =$ ____________

(b) Estimate each of these by rounding first. Write the rounded calculation as well as the answer.

(iv) -6.2×3.9 is about _________________ = ___________

(v) $62 \div -7.8$ is about _________________ = ___________

A5 Word problems

Read each one **all the way to the end** before you start writing.

1. **The monthly fee.** Mr Bowen's bank takes a 4-dollar fee out of his account at the end of every month. He forgets about it for 7 months. What is the total change to his balance?

2. **The dive.** A submarine starts at the surface and descends steadily. After 6 minutes it is at -54 m. How far does it travel each minute?

3. The market stall. Mr Bowen takes 47 identical rubber ducks to the market to sell. He plans to sell them for 3 dollars each, and the market charges him a 2-dollar fee on every duck he sells, plus 15 dollars for the stall. It rains all day and nobody comes, so he sells 0 ducks. How much money does he make or lose that day?

Section B · Mathematics 1.3 — Multiples and the LCM

Remember from the lesson

A **multiple** is the answer when you multiply a number by 1, 2, 3, 4, . . . The multiples of 4 are 4, 8, 12, 16, 20, . . .

In everyday English, **common** means *ordinary*. In maths it means **shared**. A **common multiple** of two numbers is in **both** lists, and the **lowest common multiple (LCM)** is the smallest of them. You already know this idea in Vietnamese as **BCNN**.

B1 Build it from the lists

(a) Write the first six multiples of **8**.

____

(b) Write the first six multiples of **12**.

____

(c) Which numbers appear in **both** lists? These are the common multiples.

____

(d) So the **LCM** of 8 and 12 is ____________

B2 Do not just multiply them

The trap in this topic

The LCM is usually **smaller** than the two numbers multiplied together. If one number **divides into** the other, the LCM is just the **bigger** number. When you are not sure, **list them**.

Find the LCM of each pair.

(a) 3 and 7 LCM = ________

(b) 5 and 20 LCM = ________

(c) 6 and 8 LCM = ________

(d) 9 and 9 LCM = ________

(e) A student says the LCM of 5 and 20 is 100, because $5 \times 20 = 100$. In one full sentence, explain what they have got wrong.

B3 Word problems

1. **The two lighthouses.** One lighthouse flashes every 6 seconds. The other flashes every 10 seconds. They have just flashed at the same moment. How many seconds until they next flash together?

2. **The bread and the cheese.** Mr Bowen is making sandwiches. Bread rolls come in packs of 8. Slices of cheese come in packs of 12. He wants one slice of cheese in every roll, with nothing left over. What is the smallest number of sandwiches he can make, and how many packs of each must he buy?

B4 Now you write the question

Your turn to be the teacher

So far the questions have given you the English, and you have written the maths. Now do it **backwards**. You are given the maths — write the English.

Your word problem must:

- be about something **real**;
- be written in **full English sentences**, and end with a question;
- have the answer that is already given to you.

(a) Write a word problem for $7 \times -4 = -28$

(b) Write a word problem for **LCM of 4 and 10** = 20

This one must be about two things that keep repeating, and the question must ask when they next happen together.

Section C · Science 1.2 — Comparing Plant and Animal Cells

Remember from the lesson

An animal cell has a **cell membrane**, **cytoplasm**, **mitochondria** and a **nucleus**. A plant cell has all four of those **plus** a **cell wall**, **chloroplasts** and a **sap vacuole**.

The one clue that always works is the **cell wall** — a straight, stiff edge, with the cells packed together like bricks. **"Not green" does not mean animal.**

C1 Match the word to its meaning

Write the correct letter in the box next to each word.

Word	Letter	Meaning
1. Stain	☐	**A** The job that a cell does.
2. Similar	☐	**B** The tough substance a cell wall is made of.
3. Function	☐	**C** A coloured dye added to a specimen to make its parts easier to see.
4. Specialised	☐	**D** The green substance inside a chloroplast.
5. Cellulose	☐	**E** Alike in some ways and different in others — **not** the same.
6. Chlorophyll	☐	**F** The cell has a structure that helps it do its job really well.

C2 Plant or animal? And how do you know?

Mr Bowen looks at four slides down the microscope. For each one, write **plant** or **animal**, and then give the **one feature** that tells you.

	What Mr Bowen sees	Plant or animal?	The feature that tells you
1	Cells with straight, stiff edges, packed together like bricks, and full of small green discs.		
2	Round, separate cells with no straight edges anywhere.		
3	Purple cells with no green discs at all — but every cell has straight sides.		
4	Long, soft fibres with many dark nuclei, and no straight edges.		

One of these four catches almost everybody. Read the orange box at the top of this section again before you answer.

C3 Similar is not the same

Word help — three patterns for comparing

"A and B are **similar because** both of them …"
"They are **different because** A has …, **but** B …"
"**Unlike** an animal cell, a plant cell …"

1. Finish this sentence: "A plant cell and an animal cell are **similar because** both of them …"

2. Finish this one: “They are **different because** a plant cell has . . . , **but** an animal cell . . . ”

3. An animal cell has no fixed shape. Explain why, in one full sentence.

C4 Put the method in order

These are the steps for making a stained slide of your own cheek cells, but they are in the wrong order. Number them **1** to **6**.

☐	Add one drop of methylene blue stain to the sample.
☐	Looking **from the side**, move the lens down close to the slide.
☐	Rub the inside of your cheek gently with a clean cotton bud.
☐	Look down the eyepiece and turn the focus so the lens moves **away**, until the cells are sharp.
☐	Lower a cover slip onto the drop slowly, so that no air bubbles are trapped.
☐	Wipe the cotton bud onto the middle of a clean glass slide.

Why must you look **from the side** in one of those steps? Answer in one full sentence.

Section D · Science 1.3 — Specialised Cells

Remember from the lesson

A cell’s **function** is the job it does. A **specialised** or **adapted** cell has a structure that helps it do that job really well.

The sentence to use every time: A [cell] is **adapted to** [its job] **because it has** [its special feature].

D1 Finish the table

The first row is done for you. Fill in the other four.

Name	Function (its job)	Special structure	How this helps
Red blood cell	Carries oxygen around the body.	Full of haemoglobin, and no nucleus.	Haemoglobin holds the oxygen, and losing the nucleus leaves more room for it.
Neurone			
Ciliated cell			
Root hair cell			
Palisade cell			

D2 Use the sentence, and think

1. Write the full sentence for the **root hair cell**, using **adapted to** and **because it has**.

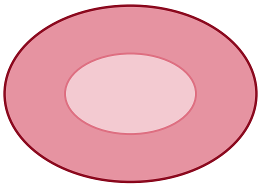
2. Write the full sentence for the **palisade cell**.

3. A red blood cell, a neurone and a ciliated cell are all animal cells. Name **two** things that all three of them have.
   Careful. Check each one before you write.

4. Root hair cells have **no** chloroplasts. Suggest why not.

D3 Name the cell

Each drawing is one of the five specialised cells. Write its name and its job in the table below.



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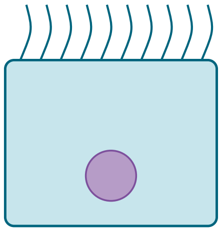

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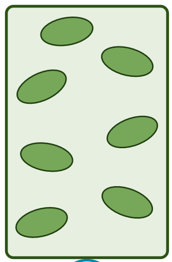

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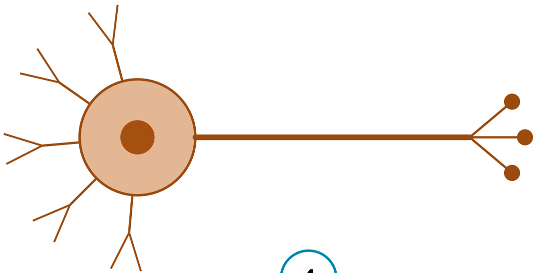

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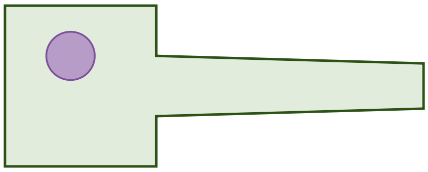

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No.	Name of the cell	Its job
1		
2		
3		
4		
5		